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PS E:\Decode AI Internship\Project 2\project-2-data-classification> python -m venv .venv
PS E:\Decode AI Internship\Project 2\project-2-data-classification> .venv\Scripts\activate
(.venv) PS E:\Decode AI Internship\Project 2\project-2-data-classification> pip install pandas scikit-learn
Collecting pandas
  Downloading pandas-2.3.3-cp310-cp310-win_amd64.whl (11.3 MB)
    |████████████████████| 11.3 MB 407 bytes/s
Collecting scikit-learn
  Downloading scikit_learn-1.7.2-cp310-cp310-win_amd64.whl (8.9 MB)
    |████████████████████| 8.9 MB 22 kB/s
Collecting python-dateutil>=2.8.2
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
    |████████████████████| 229 kB 2.2 MB/s
Collecting numpy>=1.22.4
  Downloading numpy-2.2.6-cp310-cp310-win_amd64.whl (12.9 MB)
    |████████████████████| 12.9 MB 5.1 kB/s
Collecting tzdata>=2022.7
  Downloading tzdata-2026.2-py2.py3-none-any.whl (349 kB)
    |████████████████████| 349 kB 467 kB/s
Collecting pytz>=2020.1
  Downloading pytz-2026.2-py2.py3-none-any.whl (510 kB)
    |████████████████████| 510 kB 233 kB/s
Collecting threadpoolctl>=3.1.0
  Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Collecting joblib>=1.2.0
  Downloading joblib-1.5.3-py3-none-any.whl (309 kB)
    |████████████████████| 309 kB 939 kB/s
Collecting scipy>=1.8.0
  Downloading scipy-1.15.3-cp310-cp310-win_amd64.whl (41.3 MB)
    |████████████████████| 41.3 MB 4.6 kB/s
Collecting six>=1.5
  Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Installing collected packages: six, numpy, tzdata, threadpoolctl, scipy, pytz, python-dateutil, joblib, scikit-learn, pandas
Successfully installed joblib-1.5.3 numpy-2.2.6 pandas-2.3.3 python-dateutil-2.9.0.post0 pytz-2026.2 scikit-learn-1.7.2 scipy-1.15.3 six-1.1

```

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(.venv) PS E:\Decode AI Internship\Project 2\project-2-data-classification> python main.py

```

First 5 rows:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

Dataset information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	sepal length (cm)	150 non-null	float64
1	sepal width (cm)	150 non-null	float64
2	petal length (cm)	150 non-null	float64
3	petal width (cm)	150 non-null	float64
4	target	150 non-null	int64

dtypes: float64(4), int64(1)

memory usage: 6.0 KB

None

Missing values:

sepal length (cm)	0
sepal width (cm)	0
petal length (cm)	0
petal width (cm)	0
target	0

dtype: int64

Accuracy: 1.0

Accuracy: 1.0

Classification Report:					
		precision	recall	f1-score	support
0		1.00	1.00	1.00	10
1		1.00	1.00	1.00	9
2		1.00	1.00	1.00	11
accuracy				1.00	30
macro avg				1.00	30
weighted avg				1.00	30

Confusion Matrix:

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Model saved successfully in models/iris_model.pkl

(.venv) PS E:\Decode AI Internship\Project 2\project-2-data-classification> python predict.py

Flower Prediction Result

Sepal Length : 5.1 cm

Sepal Width : 3.5 cm

Petal Length : 1.4 cm

Petal Width : 0.2 cm

Predicted Flower Class: 0

Predicted Flower Name : Setosa

Prediction completed successfully.

(.venv) PS E:\Decode AI Internship\Project 2\project-2-data-classification> █